

HANZLA SIBGHAT

Software Engineer

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WORK EXPERIENCE

Siemens AG

Software Engineer

Munich, Germany

Feb 2025 - Present

- Acted as the lead AI engineer of the **SieSight** browser extension, one of Siemens' most popular, global AI tools for internal employees (> 10,000 users)
- Contributed to a highly accurate, reliable multi-agent architecture, including query routing, specialist agents, tool execution, MCP connectivity, asynchronous processing, and incremental failure handling
- Implemented 50+ TypeScript-based tools to enable agents to control the browser fully autonomously to automate repetitive workflows
- Implemented an enhanced **Retrieval-Augmented Generation (RAG)** pipeline with semantic retrievals, priority-based ranking, and optimized context construction for downstream LLM queries
- Built a multi-step analysis pipeline for complex, multimodal **PDF** files, resulting in Siemens' most accurate PDF parsing algorithm (confirmed by IT DA)
- Enabled scalable processing of very long, complex documents, e.g. tenders, by utilizing batch processing, offscreen rasterization, and large-payload stashing
- Securely integrated existing Siemens AI platforms such as **SiemensGPT**, code.siemens.com open-source models, and ai-attack OpenAI models, with model-specific configuration, authentication, and request throttling
- Ensured high code quality by continuously debugging, linting, and adding automated tests incl. code test coverage quality gates
- Technologies: LangGraph, LangChain, React, TypeScript, Chrome/Edge browser extension development, Jest, Playwright*

National Database and Registration Authority (NADRA)

Software Engineer

Islamabad, Pakistan

Feb 2024 - Sept 2024

- Built end-to-end full-stack features for several critical national identity workflows
- Developed multi-step citizen-facing UI flows with form validation, real-time feedback, and responsive layouts for kiosk and web use
- Implemented and consumed **APIs** with key-value database caching and database query improvements, reducing average response latency by about **50%**
- Worked with real-time async messaging between services and wrote both unit and integration tests
- Technologies: Next.js, React, Java, Spring Boot, REST APIs, Redis, Kafka/RabbitMQ, JUnit, Mockito, and Jest*

PLC Group

Software Engineer

Ontario, Canada (Remote)

July 2023 - Feb 2024

- Worked as a full-stack developer on a telecom monitoring platform, building UI screens in and connecting them to APIs
- Implemented dashboard components, charts, and client-state management for cleaner data loading and updates
- Helped improve page load and API response times by about **50%** through caching, query batching, and frontend performance fixes, and contributed to testing and code reviews
- Technologies: React, TypeScript, Node.js, GraphQL, Redux Toolkit, React Query*

SKILLS

Generative AI: LangGraph, LangChain, multimodal context processing, prompt optimization, Retrieval Augmented Generation (RAG)

Data & Infra: Amazon Web Services (AWS), PostgreSQL, MySQL, Redis, Kafka, RabbitMQ, Docker, CI/CD (Github Actions)

Backends & APIs: Flask, Node.js, Express.js, Spring Boot, REST, GraphQL

Frontend: React, Redux, Next.js, Material UI, web extensions (manifest V3), HTML, CSS

Tools: Claude Code, Github Copilot, Vision Studio Code, Git

Languages: TypeScript, JavaScript, Python, SQL, Java

EDUCATION

University of Passau

M.Sc. in Computer Science

Passau, Germany

Oct 2024 – Feb 2027

FAST National University of Computer and Emerging Sciences

B.Sc. in Software Engineering

Islamabad, Pakistan

Aug 2019 - Jun 2023

LANGUAGES

English (C1), **German** (A2 – Currently advancing to B1), **Urdu** (Native)